

Matching the Target Is Not Enough: A Controlled Study of Targeted Instruction Selection

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Abstract

Targeted instruction selection assumes that training subsets better aligned with a small query set should yield greater downstream utility. We introduce directional second-moment matching, an exact greedy coreset objective that matches a query set in projected-gradient geometry, and show in controlled MMLU experiments that the second-order representation improves over the first-order target representations we study. Yet stronger target matching does not reliably translate into stronger task performance: on held-out BBH, DSMC attains the closest target geometry while underperforming Random selection, a reversal observed with both Llama-2-7B and Llama-3.2-3B model stacks. On the same query examples used to define the targeting signal, target-aware methods also improve the pipeline’s operational differentiable surrogate without corresponding gains in the chain-of-thought task metric. These results show that successfully optimizing target-gradient alignment is not sufficient for downstream utility, and motivate selection objectives whose intermediate targets are validated against the quantities ultimately used to evaluate the model.

1 Introduction

Instruction tuning draws from increasingly large and heterogeneous pools, while a deployment often requires only a narrower set of capabilities. Targeted instruction selection addresses this mismatch by using a small query set to identify candidate examples that should be most useful for a target task. Recent methods operationalize usefulness through semantic retrieval, gradient similarity, influence approximations, target subspaces, or task-specific rewards (Xia et al., 2024; Kang et al., 2024; Wang et al., 2025; Min et al., 2026). Across these formulations lies an intuitive surrogate assumption: if a selected subset matches the target more faithfully, training on it should improve target utility.

This paper directly tests that assumption. Doing so requires more than observing that one selector loses to another: a weak downstream result could simply mean that the selector failed to optimize the target representation it was meant to match. We therefore construct a deliberately measurable target-matching instrument. Given unit-normalized projected gradients u , we represent a set by its directional second moment,

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \quad (1)$$

Under the quadratic kernel $k_2(u, v) = (u^\top v)^2$, empirical maximum mean discrepancy (MMD) is exactly the squared Frobenius distance between directional second moments. Our method, Directional Second-Moment Coreset (DSMC), greedily minimizes this set-level discrepancy. It consequently lets us verify whether target geometry is actually optimized rather than inferring success from downstream performance.

The experiments reveal a positive result followed by a sharper negative one. In controlled MMLU stress tests, the second-order representation improves over the first-order alternatives we examine, and at a 5% data budget DSMC consistently outperforms the targeted baselines under our shared protocol. This establishes DSMC as a useful instrument rather than a deliberately weak straw man. However, the advantage is not an advantage over target-

agnostic sampling: DSMC is indistinguishable from Random in observed mean performance at 5% and trails Random at 1%.

We then evaluate on held-out BBH queries and examples drawn from the same task family. Here the reversal becomes explicit. Across three query/selection draws, DSMC obtains the lowest second-moment discrepancy in every draw, yet loses to Random in every draw. The direction repeats on a second model stack: Llama-2-7B shows a -2.94 percentage-point DSMC–Random gap and Llama-3.2-3B a smaller -0.89 point gap. Thus the geometric objective did not merely fail to optimize; it optimized successfully without delivering utility.

Finally, we locate the break more precisely. On the same 64 query items used to define the selection signal, target-aware methods reduce the wrapped final-answer cross-entropy computed by the targeting pipeline, while chain-of-thought exact match does not improve. Query-to-test shift therefore cannot by itself explain the result. A serialization diagnostic narrows the claim: the behavior of bare-context cross-entropy differs across model stacks, so we do not claim that cross-entropy is generically misaligned with chain-of-thought evaluation. The supported conclusion concerns the exact operational surrogate used by the selection pipeline.

Contributions.

- We formulate targeted selection as matching the directional second moment of unit projected gradients and derive an exact greedy MMD coreset rule. This provides a target-matching objective whose success can be measured directly.
- We separate representation from selection in controlled MMLU experiments. Directional second moments improve the studied targeted selectors, while the extra coreset benefit is budget dependent.
- We give an instrumented empirical counterexample to the target-matching assumption. On BBH, better target-gradient geometry and a better operational query surrogate do not imply better task utility, and the direction is observed on two model stacks.
- We report negative controls and scope limits—Random and no-SFT references, a matched-format control, equal-step training, same-item diagnostics, serialization checks, and contamination audits—rather than interpreting the reversal as a uniquely identified mechanism.

2 Target Matching with Directional Second Moments

2.1 Targeted instruction selection

Let $\mathcal{C} = \{x_i\}_{i=1}^N$ be a candidate instruction pool and $\mathcal{Q} = \{q_j\}_{j=1}^M$ a small target query set. Given a budget K , the goal is to select $\mathcal{S} \subset \mathcal{C}$, $|\mathcal{S}| = K$, such that supervised fine-tuning on $\mathcal{S}$ improves held-out target performance.

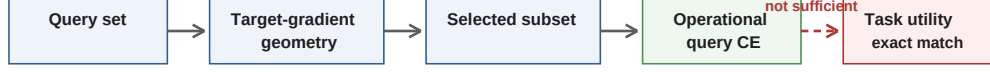
Following gradient-based targeted selection (Xia et al., 2024; Nayak et al., 2026), we compute per-example gradients at a shared warm-up checkpoint and project them to a reusable low-dimensional datastore. For an example z , let

$$u(z) = \frac{\Pi \tilde{g}(z)}{\|\Pi \tilde{g}(z)\|_2}, \quad \|u(z)\|_2 = 1, \quad (2)$$

where Π is a shared Rademacher projection. The main protocol follows the LESS-aligned asymmetry: candidate gradients are Adam-aware, while query gradients are raw SGD gradients. All statements below concern the resulting unit vectors, not unprojected parameter-space gradients.

2.2 Why a directional second moment?

First-order mean-gradient matching represents a target by $\mathbb{E}[u]$. This signed mean can cancel when useful examples occupy opposite directions, even if they share the same underlying

Targeted instruction selection assumes a surrogate chain**BBH: DSMC relative to no-SFT (three draw means)**

	Target D_2	Wrapped CE	Same-item EM	Held-out EM
Llama-2-7B	0.086 < Random 0.110	-1.16 nats	-5.21 pp	-3.33 pp
Llama-3.2-3B	0.064 < Random 0.075	-3.25 nats	-6.51 pp	-0.97 pp

Geometry and the operational surrogate improve; the task metric does not.

Figure 1: The target-matching chain and the observed break. DSMC explicitly minimizes target directional second-moment discrepancy. On BBH, it attains the lowest D_2 in every query draw on both model stacks, and target-aware methods improve the operational wrapped final-answer cross-entropy. Task utility nevertheless does not follow. Arrows report changes relative to the corresponding no-SFT model; lower D_2 and cross-entropy are better, while higher exact match is better. Values are descriptive over three query/selection draws.

orientation. We instead use the non-centered directional second moment

$$M_P = \mathbb{E}_{u \sim P}[uu^\top]. \quad (3)$$

Because $uu^\top = (-u)(-u)^\top$, this representation captures orientation rather than sign. We deliberately call it a directional second moment, not a covariance: the vectors are normalized and the mean is not subtracted.

Consider the positive semidefinite kernel

$$k_2(u, v) = (u^\top v)^2. \quad (4)$$

Since $\langle uu^\top, vv^\top \rangle_F = (u^\top v)^2$, the biased empirical MMD between a selected set $\mathcal{S}$ and query set $\mathcal{Q}$ is exactly

$$\begin{aligned} \text{MMD}_{k_2}^2(\mathcal{S}, \mathcal{Q}) &= \left\| \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} u_s u_s^\top - \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} u_q u_q^\top \right\|_F^2 \\ &\equiv D_2(\mathcal{S}, \mathcal{Q}). \end{aligned} \quad (5)$$

The identity is algebraic: DSMC matches this empirical second moment exactly as defined, but the kernel is not characteristic and does not identify the full gradient distribution.

2.3 Exact greedy coreset construction

For a candidate x , define target relevance and accumulated selected-set similarity as

$$r_{\mathcal{Q}}(x) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} k_2(u_x, u_q), \quad r_{\mathcal{S}}(x) = \sum_{s \in \mathcal{S}} k_2(u_x, u_s). \quad (6)$$

At step $m = |\mathcal{S}|$, the exact one-step marginal for minimizing the biased MMD selects

$$x^* = \arg \max_{x \notin \mathcal{S}} \left[r_{\mathcal{Q}}(x) - \frac{r_{\mathcal{S}}(x) + k_2(u_x, u_x)/2}{m+1} \right]. \quad (7)$$

For unit vectors $k_2(u_x, u_x) = 1$. The first term rewards query alignment and the second penalizes selected-set self-similarity. We use the deterministic exact marginal throughout; we make no stochastic-greedy or submodularity guarantee.

Role in this paper. DSMC is not claimed to optimize downstream utility. Its purpose is to construct a strong, measurable intervention on target geometry. If DSMC lowers D_2 but utility fails to improve, the discrepancy cannot be attributed to the geometry objective being left unoptimized.

3 Experimental Design

3.1 Candidate pool, features, and training

The candidate pool contains 270,679 processed examples from Flan v2 (100,000), CoT (100,000), Dolly (15,011), and OASST1 (55,668), matching the pool used by LESS-style targeted selection. For each model stack, a LoRA warm-up checkpoint is used to construct a candidate-gradient datastore. We use an 8,192-dimensional Rademacher projection with seed 123, unit-normalize the projected vectors, and generate query-gradient caches separately for each draw. The selected indices are materialized into static subsets; all downstream comparisons restart from the corresponding base model rather than continuing the warm-up run.

Downstream SFT uses LoRA rank 128, alpha 512, dropout 0.05 on q/k/v/o_proj, learning rate 2×10^{-5} , linear decay, 3% warm-up, bfloat16, effective batch size 128, and four epochs. The 5% budget is $K = 13,533$ (420 optimizer steps) and the 1% budget is $K = 2,707$ (84 steps). Full configurations, prompt templates, hashes, and data-processing details are provided in Appendix G.

3.2 Target families

MMLU controlled stress test. We construct ten globally disjoint 64-example query sets: five STEM-majority draws (51 STEM, 13 humanities) and five humanities-majority draws with the proportions reversed. Evaluation is balanced accuracy over held-out MMLU STEM and humanities subjects (Hendrycks et al., 2020). Draw index is paired with SFT seed $\{42, 1, 2, 3, 4\}$; for comparisons shared across the two skew directions, the primary descriptive unit is the five direction-averaged draw-index blocks.

BBH external validation. We stratify 27 BIG-Bench Hard tasks into a query reservoir and a frozen held-out split (Suzgun et al., 2022). Three 64-example query draws define the selection signals; the held-out evaluation contains 5,209 examples and uses pinned chain-of-thought few-shot exact match. Each query/selection draw is crossed with two SFT seeds, which are averaged within the draw. The statistical unit is therefore the draw ($n = 3$), not the six seed-level runs.

3.3 Comparators and reporting policy

The controlled gradient baselines are: first-order mean-gradient TopK (LESS-style), first- and second-order relevance TopK, first- and second-order round-robin (RR), GIST whitening on the shared projected datastore (GIST-SharedProj), and a task-metric adaptation of NICE (NICE-MMLU-EM). Because several are shared-protocol adaptations rather than full official end-to-end reproductions, we use these qualified names in the paper. Target-independent controls are uniform Random- K , length-matched Random on MMLU, a sequence/label-length-matched Random control on BBH, and a shared no-SFT reference.

The study is descriptive. We report draw-level means, paired differences, and win counts; we do not report significance tests over $n = 3$ or $n = 5$ blocks. Subtask breakdowns and individual SFT seeds are diagnostics, not independent replicates.

4 Positive Control: Second Moments Help on MMLU

We first ask whether the proposed instrument contains a meaningful positive result. Table 1 factors the method into target representation (signed first order versus directional second

Table 1: Representation $\times$ selector attribution on MMLU. Balanced accuracy at one fixed seed for each skew direction. The second-order representation improves both TopK and MMD; the smaller DSMC–Second-TopK increment is within the measured training-noise scale and is not treated as a stand-alone statistical claim.

Target	1st-TopK	1st-MMD	2nd-TopK	DSMC
STEM-majority	39.77	38.47	40.68	41.10
Humanities-majority	39.31	38.24	39.92	40.54

Table 2: MMLU target-draw results. Mean balanced accuracy over ten query draws (five paired draw-index blocks). DSMC is the strongest mean targeted method at both budgets, but it shows no observed advantage over Random at 5% and trails Random at 1%. The no-SFT reference is shared, not a replicate.

Method	5%	1%
No-SFT	40.03	40.03
LESS-style TopK	39.41	39.37
First-RR	38.80	38.20
Second-RR	39.48	40.00
GIST-SharedProj	38.85	39.45
NICE-MMLU-EM	39.25	39.42
DSMC	40.36	40.17
Random- K	40.37	40.95
Length-matched Random	40.42	40.80

order) and selection rule (relevance TopK versus MMD coreset). Moving to the second-order representation improves both selectors in both skew directions: +0.61–0.91 points under TopK and +2.30–2.63 points under MMD. The selector interaction is representation dependent: first-order MMD hurts relative to first-order TopK, while second-order MMD adds 0.42–0.62 points over second-order TopK. The latter comparison is single-seed and small, so the robust conclusion is about the representation; the coreset increment is tested further across independent target draws.

Across ten target draws at the 5% budget, DSMC beats LESS-style TopK and both RR baselines in every draw, and beats the examined GIST and NICE adaptations in nine of ten draws (Table 2). Relative to Second-RR, the mean gain is 0.88 points and holds in all ten cells, supporting a complementary coreset effect at this budget. The positive claim is deliberately scoped: in the studied Llama-2 MMLU setting, directional second moments materially improve target-aware selection.

The absolute comparison creates the first reversal. Against Random- K , DSMC has a -0.01 point mean difference at 5%, winning two of five direction-averaged blocks; length-matched Random gives the same qualitative conclusion. Tightening the budget does not rescue target awareness. At 1%, the DSMC–Random gap becomes -0.77 points and no query-direction cell favours DSMC. The extra MMD gain over Second-RR also contracts to 0.17 points and five of ten cells.

The no-SFT line changes the interpretation of the targeted leaderboard. Most targeted methods fall below the base reference at both budgets. DSMC stays slightly above it, but Random is the strongest absolute method at 1%. An equal-step sensitivity repeats the 1% subsets to 420 steps; both DSMC and Random fall to approximately 38.3%, so extra optimization does not rescue the targeted subset (Appendix D).

[PENDING FINAL EXPERIMENT: Insert the pre-registered Llama-3.2 MMLU 5% result here. It will determine whether the transferable positive result is the second-order representation, the full DSMC coreset, or neither; it does not change the paper’s central BBH conclusion.]

Table 3: Held-out BBH exact match. Means first average the two SFT seeds within each of three query/selection draws. All method means are below the corresponding no-SFT reference. DSMC loses to Random in every draw on both model stacks, while achieving the lowest target second-moment discrepancy in every draw.

Stack	Method	Draw 0	Draw 1	Draw 2	Mean	Δ vs. no-SFT
Llama-2-7B	No-SFT		39.64			–
	Random- K	39.30	38.78	39.67	39.25	–0.39
	First-RR	36.24	37.61	37.61	37.15	–2.49
	Second-RR	36.80	37.00	36.80	36.87	–2.77
	DSMC	36.36	36.47	36.10	36.31	–3.33
Llama-3.2-3B	No-SFT		47.11			–
	Random- K	47.32	46.61	47.17	47.03	–0.08
	First-RR	46.11	46.52	47.04	46.56	–0.55
	Second-RR	46.62	46.23	47.18	46.68	–0.44
	DSMC	45.98	45.88	46.57	46.14	–0.97

5 External Reversal on BBH

MMLU queries are deliberately skewed, leaving open the possibility that the selector faithfully follows an unrepresentative query sample. BBH removes this specific explanation: queries and evaluation examples are held-out samples from the same task family. The result is stronger, not weaker (Table 3). On Llama-2-7B, Random reaches 39.25% micro exact match, while DSMC reaches 36.31%, a -2.94 point paired gap with zero of three draws favouring DSMC. Both RR variants also outperform DSMC; hence the MMLU claim that DSMC is the strongest targeted selector does not transfer to BBH.

The direction repeats on Llama-3.2-3B. Random remains best at 47.03%, and DSMC is last among the four shared arms at 46.14%. The -0.89 point gap is negative in all three draws. The magnitude is substantially attenuated: all methods are within one point of the no-SFT model, so the two stacks should not be presented as equally dramatic. They nevertheless agree on the load-bearing directional fact.

All SFT means are below no-SFT. On Llama-2, Random is only 0.39 points below the shared base reference, while target-aware methods are 2.49–3.63 points below it. On Llama-3.2, Random is 0.08 points below base and DSMC 0.97 below. These are shared reference lines rather than replicated treatments, so we describe the observed means rather than claim inferentially established negative transfer.

Matched-format control. On Llama-2, constraining Random to match DSMC’s coarse sequence-length $\times$ loss-bearing-label-length histogram lowers Random from 39.25% to 38.05%, but still leaves it 1.74 points above DSMC. This is consistent with instruction-format or correlated provenance composition contributing to the gap. It is not a causal decomposition: the control changes several correlated properties simultaneously. Full results are in Appendix D.

6 Where the Surrogate Chain Breaks

6.1 The target geometry is optimized

For every BBH draw, we compute the exact discrepancy optimized by DSMC,

$$D_2(\mathcal{S}, \mathcal{Q}_d) = \|M_{\mathcal{S}} - M_{\mathcal{Q}_d}\|_F^2. \quad (8)$$

On Llama-2, mean D_2 is 0.0863 for DSMC and 0.1098 for Random; on Llama-3.2 it is 0.0644 and 0.0752. More importantly, DSMC has the lowest D_2 in all three draws of both stacks. Better geometry is even associated with lower accuracy within each draw: the descriptive Spearman correlation between D_2 and accuracy is positive in every draw (lower D_2 is better).

Table 4: The BBH surrogate chain for DSMC relative to no-SFT. Δ CE and same-item exact match use the 64 query items; held-out exact match uses 5,209 examples. Lower D_2 and CE are better.

Stack	D_2 wins	Δ wrapped CE	Δ query EM	Δ held-out EM
Llama-2-7B	3/3	-1.159	-5.21	-3.33
Llama-3.2-3B	3/3	-3.255	-6.51	-0.97

The same sign remains after restricting the Llama-2 comparison to target-aware methods, although the ranking samples are too small for inferential claims.

This rules out the narrow explanation that DSMC loses because its objective was not actually optimized. In MMLU, DSMC is also closer than Random to a leave-one-draw-out balanced reference in all ten draws at both budgets (Appendix C). Thus the observed failure is not simply that DSMC follows a skewed query away from the balanced reference.

6.2 The operational query surrogate improves

Gradient-based targeting uses a differentiable final-answer objective to construct query gradients. We therefore recompute teacher-forced final-answer cross-entropy on the same query items, under the exact wrapped serialization used by each stack’s targeting pipeline. On Llama-2, all four target-aware methods reduce this loss by 0.68–1.40 nats, while Random increases it by 0.33. On Llama-3.2, the target-aware methods reduce it by 3.19–3.40 nats and Random by only 1.37. The second stack therefore gives a magnitude split rather than a sign split; in both, the targeted methods move the operational surrogate more.

6.3 Task utility does not follow on the same examples

Comparing query loss to held-out exact match changes both metric and examples. We consequently evaluate chain-of-thought exact match on the same 64 query items. For Llama-2, every target-aware method’s mean query exact match falls relative to base (−3.65 to −6.51 points) despite the substantial cross-entropy reductions. On Llama-3.2, all three targeted methods again have negative mean changes, with DSMC falling by 6.51 points; Random changes by +1.30 points. The individual 64-item draw metrics are coarse—one item is 1.56 points—but the mean separation is consistent with the held-out ranking.

The same-item result matters logically. A pure account in which targeted selection improves the query examples but fails to generalize cannot explain the observation, because the task metric does not improve on the query examples themselves. It does not rule out every form of overfitting, nor does it identify the missing mechanism.

6.4 Serialization narrows the claim

The operational query loss uses each stack’s LlamaFactory wrapper, whereas BBH generation uses a pinned bare evaluation context. Recomputing final-answer cross-entropy on that bare context yields different behavior across stacks. On Llama-2, no method improves bare-context CE, although targeted methods degrade it much less than Random. On Llama-3.2, target-aware methods do improve bare-context CE while Random does not. Therefore we do not claim that final-answer cross-entropy is inherently misaligned with chain-of-thought generation. The claim is narrower and operational: the surrogate actually used by the targeting pipeline can improve on the defining examples without a corresponding improvement in the task metric, and the role of serialization is model-stack dependent.

7 Related Work

Instruction data selection. General-purpose instruction selection commonly scores data for quality, difficulty, diversity, or representativeness. DEITA combines complexity and diversity for data-efficient alignment, while principled warm-up selection studies how small subsets

affect later fine-tuning (Liu et al., 2024; Kang et al., 2024). Our focus is narrower: selection for a specific target described by a small query set.

Targeted and influence-based selection. LESS builds reusable low-dimensional gradient datastores and uses optimizer-aware gradient similarity for targeted instruction tuning (Xia et al., 2024). NICE replaces next-token-loss influence with a policy-gradient construction tied to non-differentiable evaluation metrics (Wang et al., 2025). GIST recovers a target-gradient subspace and scores candidate alignment under coupled optimization geometry (Min et al., 2026). A recent controlled study separates representations from selection algorithms and finds that gradient representations are often predictive, but rankings vary across tasks, models, and budgets (Nayak et al., 2026). We differ by directly optimizing a measurable set-level geometry and then verifying that the optimized geometry itself fails to imply utility.

Surrogate mismatch and strong Random baselines. ROSE motivates reward-oriented selection by documenting non-monotonicity between instruction-tuning loss and task reward (Wu et al., 2025); NICE makes a related move from differentiable training loss to task metrics (Wang et al., 2025). Large-scale studies also find that sophisticated selectors can be difficult to distinguish from or can underperform Random (Xia et al., 2025). Our contribution is not another broad selector leaderboard and not a replacement objective. It is a diagnostically resolved counterexample chain: target geometry is measurably optimized, the operational query surrogate moves in the intended direction, and task utility still does not follow.

8 Limitations

The study covers two model stacks (Llama-2-7B and Llama-3.2-3B), one candidate pool, and two target families. A model stack changes model, tokenizer, and serialization together; it is not an architecture-only ablation. BBH has only three independent query/selection draws, and the effect on the 3B stack is small. MMLU couples draw index to SFT seed, so direction-averaged blocks rather than ten cells carry comparisons shared across directions. Several baselines are controlled adaptations on a common datastore, not full official end-to-end reproductions. The contamination audit includes exact, fuzzy, and long n-gram lexical checks but not a semantic nearest-neighbor audit. Finally, the diagnostics do not identify a unique mechanism. Format and provenance are plausible contributors, while a simple task-frequency specialization account is unsupported, but neither conclusion is causal.

9 Conclusion

Within these limits, the evidence rejects a common sufficiency assumption. We made target matching stronger, verified that the selected sets were closer under the optimized geometry, and found that downstream utility still did not follow. Target-aware selectors can accomplish their intermediate objective without accomplishing the task objective. Progress in targeted instruction selection therefore requires not only better optimization of a chosen surrogate, but stronger validation that the surrogate mediates the utility we ultimately care about.

AI use statement

Generative AI tools were used to summarize and cross-check existing literature, inspect repository artifacts, suggest paper structure and wording, and draft and edit parts of the manuscript. Earlier in the research workflow, generative AI tools were also used to provide feedback on experimental design, interpretation, and code. The authors reviewed all AI-assisted text and claims against the underlying code, configurations, manifests, and result files; all reported numerical results were taken from experiment artifacts rather than generated by an AI system. The authors take responsibility for the final content, including all AI-assisted material.

Ethics statement

This work studies data selection for publicly available language-model benchmarks and does not involve human-subject data collection. The main risks are those shared by instruction tuning, including inherited dataset biases, benchmark contamination, and overgeneralization from a limited experimental scope. We report negative results, adapted-baseline scope, and contamination audit limits explicitly to reduce the risk of overstating practical conclusions.

Reproducibility statement

Appendix G documents the data composition, target-draw construction, feature protocol, selection rules, training hyperparameters, evaluation pins, statistical units, and integrity checks. The supplementary material will include anonymized code, exact run configurations, frozen manifests and hashes, paper tables generated from committed result files, and the pre-registered protocols for the BBH and second-model experiments.

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A Method Derivations

A.1 MMD identity

Let $\widehat{M}_S = |\mathcal{S}|^{-1} \sum_{s \in \mathcal{S}} u_s u_s^\top$ and define $\widehat{M}_Q$ analogously. Expanding the squared Frobenius norm gives

$$\begin{aligned} \|\widehat{M}_S - \widehat{M}_Q\|_F^2 &= \frac{1}{|\mathcal{S}|^2} \sum_{s, s'} \langle u_s u_s^\top, u_{s'} u_{s'}^\top \rangle_F \\ &\quad + \frac{1}{|\mathcal{Q}|^2} \sum_{q, q'} \langle u_q u_q^\top, u_{q'} u_{q'}^\top \rangle_F \\ &\quad - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s, q} \langle u_s u_s^\top, u_q u_q^\top \rangle_F \end{aligned} \quad (9)$$

$$= \frac{1}{|\mathcal{S}|^2} \sum_{s, s'} (u_s^\top u_{s'})^2 + \frac{1}{|\mathcal{Q}|^2} \sum_{q, q'} (u_q^\top u_{q'})^2 - \frac{2}{|\mathcal{S}||\mathcal{Q}|} \sum_{s, q} (u_s^\top u_q)^2. \quad (10)$$

This is the biased empirical MMD under k_2 .

A.2 Greedy marginal

Terms depending only on $\mathcal{Q}$ can be dropped. Comparing the objective after adding x to a set of size m yields Equation 7 after positive rescaling. The implementation maintains $r_S(x)$ incrementally and masks selected or invalid rows. Zero and non-finite feature rows are rejected before selection.

B Additional MMLU Results

[TODO: Insert full ten-draw tables, target-weighted metrics, 2×2 raw subscores, the completed Llama-3.2 MMLU result, and the pre-registered outcome interpretation.]

C Additional Geometry Diagnostics

[TODO: Insert per-draw D_2 , D_2 to the leave-one-draw-out balanced MMLU reference, and descriptive rank correlations.]

D Sensitivity and Falsification Checks

[TODO: Insert the 1% equal-step arm, Random length matching, BBH sequence×label matching, task-exposure diagnostic, and bare-context CE tables. Preserve the explicit statement that these checks do not identify a unique mechanism.]

E Baseline Implementations

[TODO: Document the exact differences between the shared-protocol adaptations and official LESS, GIST, and NICE; include fidelity and numerical-validation checks.]

F Contamination and Prompt Audits

[TODO: Insert exact/fuzzy/n-gram lexical contamination results, BBH few-shot leakage audit, prompt parity, token truncation, and evaluation coverage gates. State that semantic nearest-neighbor auditing was not performed.]

G Reproducibility Details

[TODO: Insert resolved run provenance, candidate/target hashes, selection seeds, optimizer-step counts, all environment and evaluation pins, runtime and hardware summaries, and an anonymized artifact manifest.]